Heteroencoding

Baseline Experiment

This circuit setup applies additive and subtractive (inhibitory) noise to the input data. The “heteroencoder” component then clusters the noisy data, generating a unique SHR per cluster. Finally, the “binning” decoder auto-classifies the resulting SHRs.

Dataset: 10,000 random integers from 0 to 9, encoded as random SHRs with population P=10.

Outcome: The noisy test data is successfully clustered into 10 different categories.

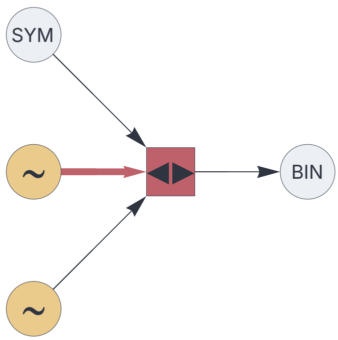
Requirements

```
<< "Circuits.m"
```

Circuit


```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",

  "hyperparameters": {"11": [1000, 30], "12": [1000, 30], "21": [1000, 10], "101": [500, 5]},

  "dataflow": [
    {"component": "noise", "send": [11]},

    {"component": "noise", "send": [12]},

    {"component": "input", "plugin": "codec", "send": [21]},

    {"component": "heteroencoder", "send": [101], "receive": [[11, {"slot": 12, "tags": ["inhibit"]}, 21]]},

    {"component": "output", "plugin": "binning", "receive": [101]}
  ]
}
```

```
circuit = Circuit[json];
circuit["schematics"][ImageSize→ 180]
```



Synthetic data

```
dataset = RandomInteger[ {0, 9}, 10000];
```

Train

```
bins = circuit["function"] /@ dataset;
```

Analyze

```
SortBy[Tally[dataset], Last]
```

```
{{0, 937}, {1, 999}, {8, 1000}, {9, 1001}, {4, 1003},
 {2, 1006}, {3, 1006}, {6, 1010}, {5, 1012}, {7, 1026}}
```

```
SortBy[Tally[bins], Last]
```

```
{{3, 937}, {7, 999}, {9, 1000}, {6, 1001}, {10, 1003},
 {1, 1006}, {2, 1006}, {8, 1010}, {5, 1012}, {4, 1026}}
```

The algorithm has identified 10 clusters and correctly classified the data.